

Supplementary Information

Digital-social connectedness and healthy ageing transitions in a multiple cohort study

Supplementary Table 1. Cohort sample flow

Cohort	Person-waves	People	Waves	Lagged PW	Lagged people	Interval rows
CHARLS	96628	25873	5	70755	23474	70121
ELSA	90069	19802	9	70267	15388	69922
HRS	208674	36529	11	171918	32960	171268
KLoSA	44273	9238	6	35035	8566	35001
LASI	73408	73408	1	0	0	0
MHAS	76506	26839	5	49657	19797	39318
SHARE	327266	136843	7	190423	83352	182452

Supplementary Table 2. Alternative healthy-ageing definitions

Definition	Exposure	Studies	OR	95% CI low	95% CI high	I ²
lenient_60pct_threshold	internet_use	4	1.58	1.44	1.73	
lenient_60pct_threshold	social_participation	6	1.28	1.07	1.53	
strict_all_available_components	internet_use	4	1.34	1.30	1.38	
strict_all_available_components	social_participation	6	1.15	1.04	1.27	
without_cognition_component	internet_use	4	1.49	1.41	1.57	
without_cognition_component	social_participation	6	1.20	1.04	1.39	

Supplementary Table 3. Inverse-probability weighting for attrition

Exposure	Term	Studies	OR	95% CI low	95% CI high
internet_use	internet_use	4	1.51	1.43	1.59
social_participation	social_participation	6	1.21	1.05	1.40

Supplementary Table 4. E-value sensitivity metrics

Analysis	Contrast	Exposure/transition	OR	E-value point	E-value CI
joint_exposure	both		1.65	2.68	2.05
joint_exposure	internet_only		1.49	2.36	2.24
joint_exposure	social_only		1.15	1.56	1.09
transition	both	gain_healthy	1.68	2.74	2.07
transition	internet_only	gain_healthy	1.44	2.23	2.00
transition	social_only	gain_healthy	1.13	1.52	1.00
transition	both	maintain_healthy	1.65	2.69	1.93
transition	internet_only	maintain_healthy	1.54	2.46	2.31
transition	social_only	maintain_healthy	1.17	1.62	1.17
exposure_history	adopted	internet_use	1.24	1.79	1.56
exposure_history	persistent	internet_use	1.51	2.39	1.96
exposure_history	stopped	internet_use	1.09	1.39	1.00

exposure_history	adopted	social_participation	1.19	1.68	1.21
exposure_history	persistent	social_participation	1.32	1.98	1.23
exposure_history	stopped	social_participation	1.18	1.63	1.10
alternative_definition	lenient_60pct_threshold	internet_use	1.58	2.53	2.24
alternative_definition	lenient_60pct_threshold	social_participation	1.28	1.87	1.33
alternative_definition	strict_all_available_components	internet_use	1.34	2.02	1.93
alternative_definition	strict_all_available_components	social_participation	1.15	1.56	1.25
alternative_definition	without_cognition_component	internet_use	1.49	2.34	2.17

Supplementary Table 5. Leave-one-cohort-out analysis for joint connectedness

Omitted cohort	Contrast	Studies	OR	95% CI low	95% CI high	I ²
CHARLS	both	3	1.72	1.40	2.11	
ELSA	both	3	1.70	1.36	2.13	
HRS	both	3	1.64	1.26	2.15	
SHARE	both	3	1.53	1.40	1.66	
CHARLS	internet_only	3	1.50	1.45	1.56	
ELSA	internet_only	3	1.49	1.44	1.55	
HRS	internet_only	3	1.49	1.44	1.55	
SHARE	internet_only	3	1.48	1.39	1.58	
CHARLS	social_only	3	1.18	1.00	1.39	
ELSA	social_only	3	1.17	1.00	1.36	
HRS	social_only	3	1.16	0.98	1.39	
SHARE	social_only	3	1.08	1.04	1.12	

Supplementary Table 6. Selected subgroup pooled estimates

Subgroup	Level	Exposure	N	OR	95% CI low	95% CI high
age_group	60-69	internet_use	103095	1.57	1.51	1.63
age_group	70-79	internet_use	72136	1.59	1.52	1.66
age_group	80plus	internet_use	31017	1.46	1.35	1.57
age_group	under60	internet_use	95244	1.61	1.54	1.67
sex	female	internet_use	175034	1.61	1.57	1.66
sex	male	internet_use	126458	1.54	1.48	1.59
education	high	internet_use	59371	1.58	1.49	1.69
education	low	internet_use	106832	1.65	1.58	1.73
education	middle	internet_use	133468	1.53	1.49	1.58
rurality	rural	internet_use	96145	1.61	1.54	1.68
rurality	urban	internet_use	176526	1.58	1.54	1.63
baseline_health	healthy	internet_use	177235	1.58	1.53	1.63
baseline_health	unhealthy	internet_use	126765	1.54	1.48	1.61
age_group	60-69	social_participation	110448	1.27	1.23	1.32
age_group	70-79	social_participation	74967	1.33	1.28	1.38
age_group	80plus	social_participation	28690	1.38	1.28	1.48
age_group	under60	social_participation	102472	1.23	1.19	1.27
sex	female	social_participation	180705	1.31	1.28	1.35
sex	male	social_participation	135872	1.25	1.22	1.29
education	high	social_participation	52006	1.31	1.24	1.37

education	low	social_participation	157497	1.25	1.22	1.28
education	middle	social_participation	104544	1.33	1.29	1.38
rurality	rural	social_participation	106884	1.24	1.20	1.28
rurality	urban	social_participation	174815	1.34	1.30	1.37
baseline_health	healthy	social_participation	183695	1.29	1.26	1.32
baseline_health	unhealthy	social_participation	134859	1.28	1.24	1.32

Supplementary Table 7. Selected variable availability

Cohort	Variable	Valid N	Total N	Valid %
CHARLS	depressive_symptoms	87755	96628	90.8
CHARLS	cognition_z	87360	96628	90.4
CHARLS	frailty_index	76851	96628	79.5
CHARLS	internet_use	57051	96628	59.0
CHARLS	social_participation	91820	96628	95.0
CHARLS	healthy_aging_binary	87425	96628	90.5
ELSA	depressive_symptoms	44675	90069	49.6
ELSA	cognition_z	47722	90069	53.0
ELSA	frailty_index	26846	90069	29.8
ELSA	internet_use	31225	90069	34.7
ELSA	social_participation	38888	90069	43.2
ELSA	healthy_aging_binary	44952	90069	49.9
HRS	depressive_symptoms	194374	208674	93.1
HRS	cognition_z	162986	208674	78.1
HRS	frailty_index	208674	208674	100.0
HRS	internet_use	182776	208674	87.6
HRS	social_participation	46765	208674	22.4
HRS	healthy_aging_binary	194325	208674	93.1
KLoSA	depressive_symptoms	28852	44273	65.2
KLoSA	cognition_z	29176	44273	65.9
KLoSA	frailty_index	44265	44273	100.0
KLoSA	internet_use	0	44273	0.0
KLoSA	social_participation	43353	44273	97.9
KLoSA	healthy_aging_binary	43454	44273	98.2
LASI	depressive_symptoms	71515	73408	97.4
LASI	cognition_z	72693	73408	99.0
LASI	frailty_index	72818	73408	99.2
LASI	internet_use	72338	73408	98.5
LASI	social_participation	72339	73408	98.5
LASI	healthy_aging_binary	72377	73408	98.6
MHAS	depressive_symptoms	70644	76506	92.3
MHAS	cognition_z	70541	76506	92.2
MHAS	frailty_index	70677	76506	92.4
MHAS	internet_use	0	76506	0.0
MHAS	social_participation	44053	76506	57.6
MHAS	healthy_aging_binary	70679	76506	92.4
SHARE	depressive_symptoms	327266	327266	100.0
SHARE	cognition_z	316144	327266	96.6
SHARE	frailty_index	259718	327266	79.4
SHARE	internet_use	201043	327266	61.4
SHARE	social_participation	252251	327266	77.1

SHARE	healthy_aging_binary	326128	327266	99.7
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